

Sem I

2020(Voc)

Time : 3 hours

Full Marks : 50

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from all the Sections as directed.

Section – A

1. Choose the correct answer from the given options : $1 \times 5 = 5$

(I) What is Size of `f()` in C ?

- (a) datatype (b) Operator
(c) Function (d) Macro

(II) Which of the following statements should be used to obtain a remainder after dividing 3.14 by 2.1 ?

- (a) `Rem = 3.14 % 2.1`
(b) `Rem = modf(3.14, 2.1)`

(c) $\text{Rem} = \text{fmod}(3.14, 2.1)$

(d) $\text{Rem} = 3.14 / 2.1$

(III) Which of the following special symbol allowed in a variable name ?

(a) * (asterisk)

(b) | (pipeline)

(c) - (hyphen)

(d) _ (underscore)

(IV) Which of the following is the correct order of evaluation for the below expression ?

$z = x + y * z / 4 \% 2 - 1$

(a) $/ \% + - =$

(b) $= * / \% + -$

(c) $/ * \% - + =$

(d) $* \% / - + =$

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(V) Which of the following is not logical operator ?

(a) &

(b) & &

(c) ||

(d) !

2. Fill in the blanks : $1 \times 5 = 5$

(I) If a variable is a pointer to a structure, then _____ operator is used to access data members of the structure through the pointer variable.

(II) If the two strings are identical, then strcmp() function returns _____.

(III) It is possible to allocated a block of memory (of arbitrary size) at run-time, using the _____ function.

(IV) A float occupies 4 bytes.

(V) array uses an integer value index to access a specific element.

Section – B

There are four questions in this Section. Answer any two questions : $5 \times 2 = 10$

3. What is pointer ? How is a pointer initialized ?

4. What is a structure ? How does a structure differ from an union ?

5. Distinguish between actual argument and formal argurment in a function.

6. What is an array variable ? How does it differ from an ordinary variable ?

Section – C

There are **five** questions in this Section. Answer any **three** questions : $10 \times 3 = 30$

7. Write a C program to find factorial of the given number using user defined function.
8. Describe the various types of relational and logical operators in C with suitable examples.
9. Write a C program to find the sum of principle diagonal elements.
10. Describe the various types of loop in C language with examples.
11. A file called "STUDENTT DAT" contains information such as student roll number, name and total marks. Write a C program to create a file to store details of n students.

A
K
B



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